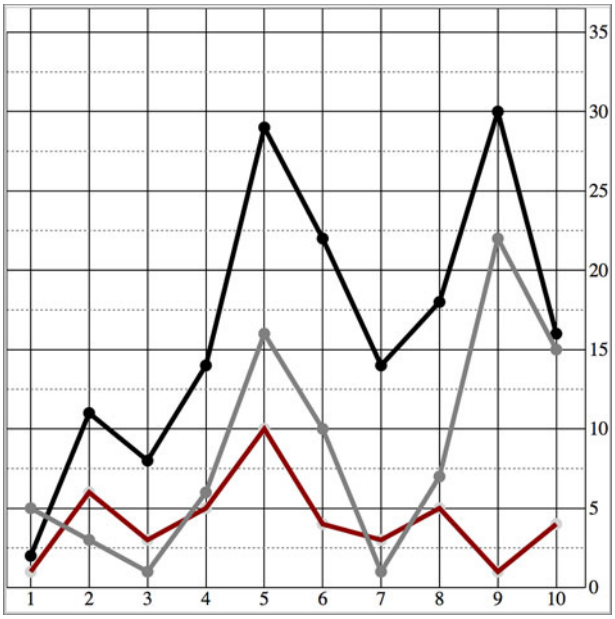


Figure 4.19 The dataset is first used to draw a set of circles, which creates the scatterplot from the beginning of this section. The dataset is then used three more times to draw each line.

4.4.3 Exploring line interpolators

D3 provides a number of interpolation methods with which to draw these lines, so that they can more accurately represent the data. In cases like tweetdata, where you have discrete points that represent data accurately and not samples, then the default "linear" method shown in figure 4.19 is appropriate. But in other cases, a different interpolation method for the lines, like the ones shown in figure 4.20, may be appropriate. Here's the same data but with the `d3.svg.line()` generator using different interpolation methods:

```
tweetLine.interpolate("basis");
retweetLine.interpolate("step");
favLine.interpolate("cardinal");
```

We can add this code right after we create our line generators and before we call them to change the interpolate method, or we can set .interpolate() as we're defining the generator.

What's the best interpolation?

Interpolation modifies the representation of data. Experiment with this drawing code to see how the different interpolation settings show different information than other interpolators. Data can be visualized in different ways, all correct from a programming perspective, and it's up to you to make sure the information you're visualizing reflects the actual phenomena.

Data visualization deals with the visual representation of statistical principles, which means it's subject to all the dangers of the misuse of statistics. The interpolation of lines is particularly vulnerable to misuse, because it changes a clunky-looking line into a smooth, "natural" line.